



# Profile

AI engineer and medical imaging researcher with publications in medical image synthesis, self-supervised medical representation learning, and anatomy-aware vision-language models. My interests center on medical image analysis, diffusion-based generation, vision-language reasoning, and reliable data/model pipelines for applied AI research.

# Education

## Sangmyung University

Mar. 2023 - Aug. 2026

### Bachelor's Program in Computer Science / Software (GPA: 3.73/4.5)

- Built a foundation in machine learning, deep learning, data preprocessing, algorithms, and software engineering.
- Applied data-driven modeling methods through preprocessing, feature engineering, validation, and model evaluation workflows.
- Developed programming and system design skills for building reliable AI and data-intensive applications.

## Publications

### **3D-LLDM: Label-Guided 3D Latent Diffusion Model for Improving High-Resolution Synthetic MR Imaging in Hepatic Structure Segmentation**

K. Kim, J. Bae, Y. Han, J. Y. Bae, S. Ju, **J. Lim**, G. Kim, N. J. Kim, W. K. Jeong, et al.

IEEE International Symposium on Biomedical Imaging (ISBI), 1-5 | 2026

### **MATHENA: Mamba-based Architectural Tooth Hierarchical Estimator and Holistic Evaluation Network for Anatomy**

K. Kim, J. Park, Y. Han, A. Jung, S. Park, S. Lee, J. Yang, J. Han, S. Lee, **J. Lim**, et al.

arXiv preprint arXiv:2604.00537 | 2026

### **MAESIL: Masked Autoencoder for Enhanced Self-supervised Medical Image Learning**

K. Kim, H. Jung, Y. Han, **J. Lim**, Y. Jean, S. Park, E. Choi, H. Go, S. Ju, S. Park, et al.

International Conference on Electronics, Information, and Communication | 2026

### **NEMESIS: Noise-suppressed Efficient MAE with Enhanced Superpatch Integration Strategy**

K. Kim, H. Jung, Y. Han, H. Go, E. Choi, S. Park, **J. Lim**, J. Yang, S. Lee, et al.

arXiv preprint arXiv:2604.01612 | 2026

### **MINTI: Fine Boundary Instructional Learning for Intelligent Anatomical Vision-Language Models**

H. Lee, Y. Han, K. Kim, **J. Lim**, I. Um, H. Go, E. Choi, K. Kim, S. Park, J. Jeong, et al.

IEIE Annual Conference, 950-953 | 2026

## Research Experience

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### Medical Imaging Lab, Seoul National University

Sep. 2025 - Present

#### Research Assistant

- Advisor: Prof. Nam-Joon Kim (SNU) and Prof. Wonjae Lee, Radiologist (Samsung Medical Center).
- Participate as a member of the SUDAL Medical Imaging Team, contributing to journal clubs, paper reviews, and collaborative research discussions.
- Built the official IMSI Lab website from the ground up to organize lab research, members, and publication records.
- Conduct research on conditional abdominal MRI synthesis using segmentation-guided ControlNet pipelines.
- Collaborate with Samsung Medical Center radiologists and NVIDIA researchers on clinically motivated medical AI projects.

## Research Collaborations

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### SUDAL Medical Imaging Team

NVIDIA Collaboration

#### Medical Imaging AI Research

- Collaborate with Prof. Nam-Joon Kim and medical AI researchers on clinically oriented medical imaging projects.
- Work with NVIDIA researcher Ping Shu Ho (Cliff) on diffusion-based medical imaging research topics.
- Focus on medical image synthesis, reconstruction, segmentation-aware generation, and robust visual representation learning.

# Key Research Projects

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## 3D Latent Diffusion for MR Imaging

2026

### Label-guided synthetic MRI and hepatic structure segmentation

- Contributed to the experimental workflow for label-guided 3D latent diffusion models for high-resolution synthetic MR imaging.
- Supported segmentation-aware generation experiments aimed at improving hepatic structure segmentation under limited medical imaging data.
- Reviewed model outputs and failure cases to connect synthesis quality with downstream segmentation utility.

## Medical Foundation Models with MAE and Mamba

2026

### Self-supervised representation learning and anatomical structure estimation

- Worked on MAE-based medical image representation learning pipelines for robust downstream evaluation.
- Explored Mamba-based modeling approaches for hierarchical anatomical estimation and efficient visual sequence modeling.
- Participated in literature review, experiment organization, and result analysis for medical foundation model research.

## Medical Vision-Language Model Development

2026

### Instruction tuning for anatomy-aware VLMs

- Developed research workflows for fine-grained boundary instruction learning in anatomical vision-language models.
- Connected medical imaging tasks with multimodal reasoning, report-aligned supervision, and clinical visual understanding.
- Contributed to dataset/task formulation for evaluating anatomy-aware vision-language reasoning.

# Professional Experience

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## INTWEEN

Jun. 2026 - Present

### AI SaaS / Data Systems Engineer

- Develop in-house AI-based SaaS products, including Sasure, by integrating internal datasets and service workflows.
- Design pipelines for collecting, cleaning, normalizing, and integrating fragmented operational data into analysis-ready formats.
- Build workflow automation systems that connect data preprocessing, retrieval, and AI-assisted decision support.
- Translate heterogeneous operational data into structured resources for enterprise AI transformation and internal model-driven services.

## GDSK

Jan. 2025 - Aug. 2025

### Data / AI Engineer

- Designed ETL processes that parse unstructured address text and convert it into standardized address formats.
- Built database architecture and preprocessing workflows for high-volume geographic and address-based datasets.
- Integrated public and geospatial datasets to construct feature sets for regional income estimation models.
- Developed data collection and preprocessing pipelines for large-scale geospatial information analysis.

## OUTTA

Jul. 2024 - Present

### Medical Imaging Team Lead

- Contribute to an AI organization connected to Seoul National University research and education communities.
- Design bootcamp curriculum and hands-on learning materials for Python, deep learning, and medical AI topics.
- Lead structured paper study sessions on medical image analysis, diffusion models, MAE-based representation learning, and VLMs.
- Coordinate research discussions and project execution across junior members to strengthen reproducible AI research practice.

# Certifications & Licenses

**TOEIC**

935

Oct. 2022

**Big Data Analysis Engineer**

Certification

Acquired

**SQLD**

SQL Developer

Acquired

**AWS SAA**

Solutions Architect Associate

Acquired

**NCA**

Naver Cloud Platform

Acquired

**Information Processing Engineer**

National Technical Qualification

Acquired

# Technical Skills

**Programming Languages:** Python, SQL, JavaScript, TypeScript, Java, Bash

**AI / ML Frameworks:** PyTorch, MONAI, Hugging Face Transformers, scikit-learn, TensorFlow

**Medical AI:** Medical Image Analysis, 3D Latent Diffusion, Masked Autoencoders, Mamba, Vision-Language Models, Segmentation-aware Generation

**LLM / LLMOps:** LLM Fine-tuning, RAG, Prompt Engineering, Model Evaluation, Workflow Automation

**Data Engineering:** ETL Pipelines, Data Cleaning, Geospatial Data Processing, Feature Engineering, Database Architecture

**Backend & Infrastructure:** FastAPI, Node.js, REST APIs, Docker, Linux, AWS, Naver Cloud Platform

**Tools:** Git, GitHub, VS Code, PyCharm, Docker Compose, Notion, Figma

# Contact Information